PROJECT REPORT: AGENTIC AI COURSE ASSISTANT

Domain: Course Assistant

Project Title: Intelligent Agentic Assistant for 13-Day AI Curriculum

Prepared by: Ankan Chatterjee, Roll No. 23051814

Batch: Agentic AI-2027

1. INTRODUCTION

In the rapidly evolving landscape of Artificial Intelligence, we are witnessing a paradigm shift from static Large Language Models (LLMs) to autonomous agents. While traditional LLMs are limited to predicting the next token based on a prompt, "Agentic AI" involves models that can reason, use external tools, and iterate on their own outputs to achieve complex goals.

This project focuses on the development of the **Agentic AI Course Assistant**, a sophisticated Retrieval-Augmented Generation (RAG) system. Unlike standard chatbots that provide generic answers, this assistant is specifically engineered to support students through a rigorous 13-day journey. It provides session-specific guidance, deep code explanations, and conceptual clarifications by acting as an intelligent bridge between complex framework documentation—such as LangChain, CrewAI, and LangGraph—and the specific pedagogical flow of the course curriculum.

2. PROBLEM STATEMENT

The Challenge

Students participating in the **Agentic AI Course** are exposed to a high-density curriculum that evolves rapidly over 13 days. The materials cover a wide spectrum of the agentic ecosystem:

- **Day 1-2:** Foundational LLM APIs and the transition to Tool Calling / Function Calling.
- **Day 3-5:** Managing state with Memory Systems and deep dives into the LangChain framework.
- **Day 6-7:** Orchestrating multi-agent collaboration and role-playing using CrewAI.
- **Day 8-10:** Building advanced graph-based workflows with LangGraph and implementing Autonomous RAG.
- **Day 11-13:** Quality assurance via Evaluation, Deployment strategies, and final Capstone projects.

Student Pain Points

1. **Information Overload:** With 13 days of distinct yet interconnected content, students often struggle to recall where specific concepts were introduced. For example, distinguishing between a "Basic Tool" in Day 2 and a "LangGraph Node" in Day 8 can be challenging without immediate context.
2. **Context Fragmentation:** Students often find it difficult to relate early fundamental concepts (like simple Tool Calling) to advanced, stateful implementations (like LangGraph State Management). This lack of "connective tissue" can hinder the mastery of complex agentic workflows.
3. **Debugging Obstacles:** The course relies heavily on Jupyter notebooks (.ipynb).

When code fails, students need more than just an error message; they require an explanation of the underlying logic and how it relates to the course-specific implementation of that day.

4. **Navigation Friction:** Manually searching through 13 separate notebooks, various JSON knowledge bases, and slide decks is inefficient and disrupts the learning "flow," leading to frustration and slower progress.

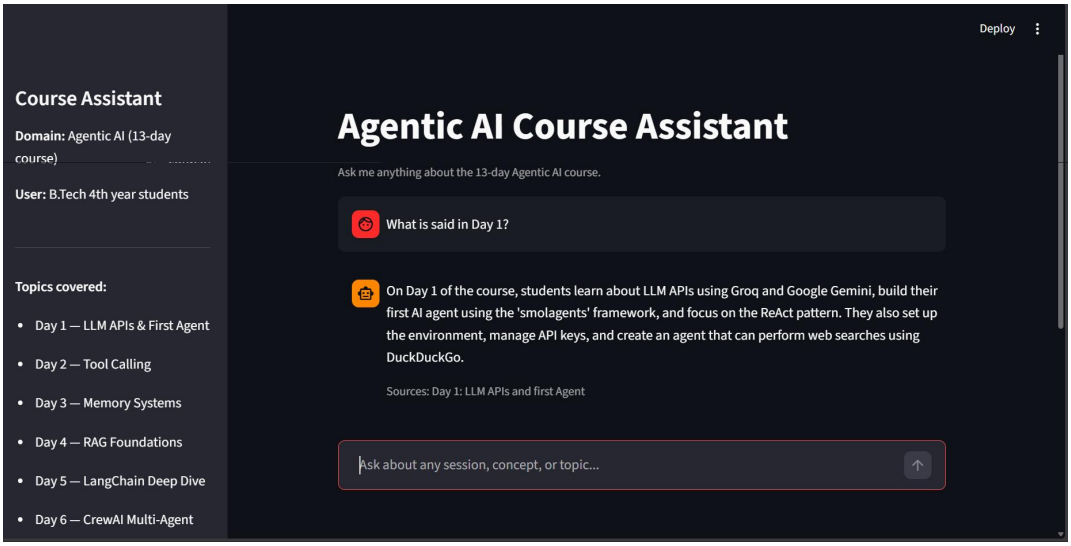
3. SOLUTION & FEATURES

To address these challenges, we built an agentic solution that combines deep retrieval with advanced reasoning capabilities to act as a 24/7 personalized tutor.

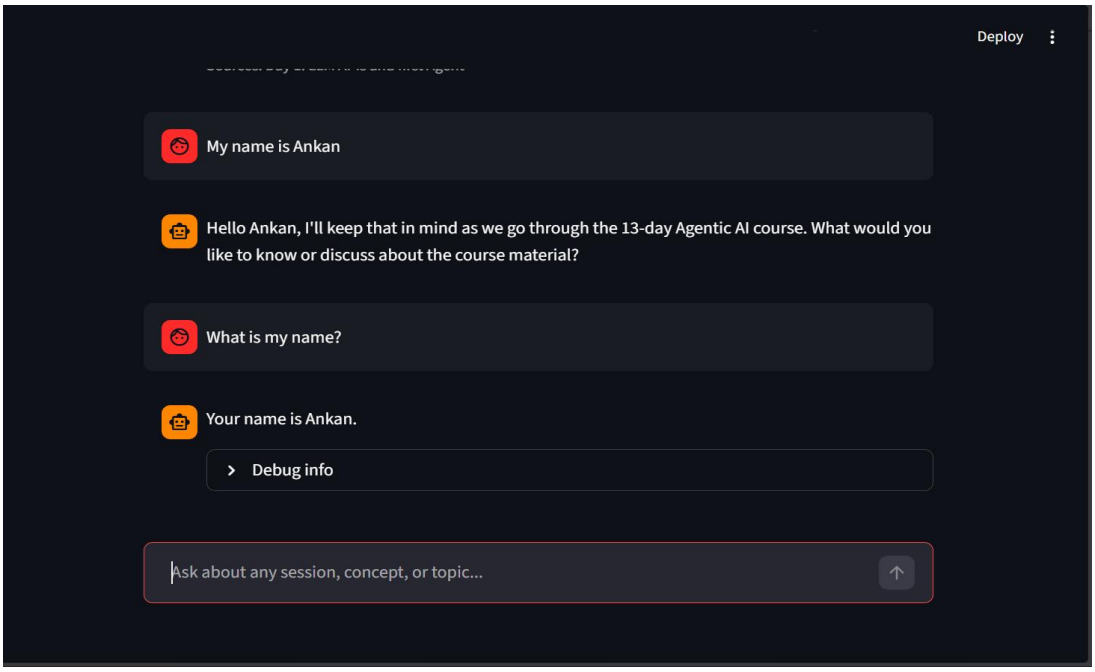
Core Features

- **Session-Aware Retrieval:** The system indexes content with temporal metadata from all 13 days. When a student asks a question, the assistant doesn't just find a match; it identifies the *specific session* relevant to the query, providing answers that respect the learning stage of the student.
- **Intelligent RAG (Retrieval-Augmented Generation):** By utilizing the custom knowledge_base.py and structured doc.json, the assistant ensures high-fidelity responses. It prioritizes course-specific terminology and methodology over general web knowledge, ensuring consistency with the instructor's lessons.
- **Code Explanation Engine:** The assistant is capable of parsing and interpreting logic directly from the 13 .ipynb notebooks. It can take a snippet of code and explain its purpose within the context of that day's objective—for instance, explaining how a specific AgentState is updated in a LangGraph node.
- **Agentic Reasoning Loop:** Using a "Plan-and-Execute" or "Reason-then-Act" loop, the assistant evaluates the student's query to determine if the retrieved information is sufficient. If the retrieval is ambiguous, the agent is programmed to "self-correct" by broadening its search or asking the student for more context.

4. SCREENSHOTS



Caption: The primary dashboard where students interact with the Agentic Assistant, featuring the sidebar for session selection and the main chat window.



Caption: Depiction of the agent processing and retaining conversational memory within an interaction

5. TECH STACK

The project utilizes a state-of-the-art AI stack designed for reliability, modularity, and rapid response:

- **Orchestration Framework: LangChain** serves as the backbone, managing the complex chaining of LLM prompts and the integration of various agentic tools.
- **Agentic Logic: LangGraph and CrewAI** are employed for more complex, non-linear workflows, allowing the assistant to mimic the multi-agent collaboration taught in the later sessions of the course.
- **Reasoning Models: llama-3.3-70b-versatile** acts as the primary "brain" for high-level reasoning and final response synthesis.
- **Vector Database: ChromaDB** provide the storage for high-dimensional embeddings. These databases allow for semantic search across the 13-day curriculum with sub-second retrieval times.
- **Frontend Interface: Streamlit** was chosen for its ability to quickly turn Python scripts into interactive web apps, making the assistant easily accessible to students without complex setup.
- **Backend & Data Processing: Python** handles the core logic, while **JSON and Jupyter Notebooks** serve as the structured and unstructured source materials that form the assistant's knowledge base.

6. UNIQUE POINTS

What makes this assistant truly "Agentic" and sets it apart from generic AI tutors:

1. **Temporal Context Awareness:** The assistant possesses an architectural understanding of the course's chronological hierarchy. It recognizes that Day 10's "RAG with Memory" is an evolution of Day 4's "Basic RAG," allowing it to explain the "why" behind the added complexity.
2. **Notebook-to-Knowledge Mapping:** It creates a direct semantic link between the theoretical descriptions found in the doc.json files and the actual, executable code blocks in the notebooks. This ensures that theory and practice are always presented together.
3. **Autonomous Self-Correction:** If an initial retrieval fails to yield a high-confidence answer, the agent doesn't hallucinate. Instead, it utilizes its tool-calling capabilities to perform a recursive search through the metadata or suggests the most relevant session for the student to review manually.
4. **Zero-Hallucination Guardrails:** The system is "grounded" strictly in the course files. It is fine-tuned with system prompts that instruct the model to prioritize course materials and explicitly state "I don't know" if a topic falls outside the 13-day scope, maintaining the integrity of the educational content.

7. FUTURE IMPROVEMENTS

The roadmap for the Agentic AI Assistant aims to deepen its integration into the student's learning lifecycle:

- **Auto-Grading and Feedback Integration:** Future versions will allow the agent to review student-submitted code from the Day 13 Capstone projects, offering constructive feedback based on the best practices taught throughout the course.
- **Multi-Modal Curriculum Support:** We plan to integrate vision capabilities so the agent can "see" and explain complex architectural diagrams from the course slides, providing a more visual learning experience.
- **Specialized Collaborative Multi-Agents:** Implementing a "Multi-Agent Team" within the assistant—for example, a "Code Reviewer Agent" that focuses on syntax and a "Theory Tutor Agent" that focuses on conceptual understanding—to provide more holistic support.
- **Personalized Learning Analytics:** Integrating a database to track student queries over time. This would allow the assistant to identify individual knowledge gaps and suggest specific sessions or exercises for review, creating a truly customized learning path.

8. CONCLUSION

The **Agentic AI Course Assistant** is more than just a chatbot; it is a specialized pedagogical tool designed for the modern AI learner. By centralizing the diverse knowledge of a 13-day intensive program into a single, intelligent interface, it empowers students to master complex concepts with unprecedented efficiency. This project doesn't just teach Agentic AI—it lives it, demonstrating the practical application of the very frameworks and logic taught in the curriculum. Ultimately, it serves as a blueprint for the future of AI-assisted education, where learning is guided by agents that understand not just the "what," but the "how" and the "when" of the material.

(End of Report)